A New Auxiliary Model Approach For Fermionic Criticality

Insights on Mottness in Strongly Correlated Systems

ABHIRUP MUKHERJEE

Presubmission Open Seminar

IISER KOLKATA

Emergent Phenomena in Quantum Matter group
IISER Kolkata

EPQM

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Publications

New J. Phys. 25 113011. **AM**, N S Vidhyadhiraja, A Taraphder, S Lal. (2023)

J. Phys. A: Math. Theor. 57 275401. **AM**, S Patra, S Lal. (2024)

New J. Phys. 28 013503. **AM**, S R. Hassan, A Mukherjee, N S. Vidhyadhiraja, A Taraphder, S Lal. (2026)

in progress. **AM**, S Lal. (2026)

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in progress. **AM**, S Lal. (2026)

Phys. Rev. B 105, 085119. A Mukherjee, **AM**, N S. Vidhyadhiraja, A Taraphder, S Lal. (2022)

J. Phys.: Condens. Matter 35 315601. S Patra, **AM**, A Mukherjee, N S Vidhyadhiraja, A Taraphder, S Lal. (2023)

Phys. Rev. B 111, L140414. A Kumar N M, S Mukherjee, **AM**, A Punjal, S Purwar, T Setti, S Prabhu S., S Lal, N Kamaraju. (2025)

arXiv:2602.20600. D Debata, **AM**, S Lal. (2026)

Outline of Thesis

Chap 1. Introduction

Chap 2. Methods and Preliminaries

Chap 3. Kondo frustration via charge fluctuations: Mott localisation in $d = \infty$ [**New J. Phys. 25 113011**]

Chap 4. A New Auxiliary Model Mapping For Correlated Lattice Models [**New J. Phys. 28 013503**]

Chap 5. Mott Criticality as the Confinement Transition of a Pseudogap-Mott Metal in $d = 2$ [**New J. Phys. 28 013503**]

Chap 6. Competing tendencies In Heavy Fermion Systems: Kondo screening vs. Mott localisation [**in progress**]

Chap 7. Holographic entanglement renormalisation for fermionic quantum matter [**J. Phys. A: Math. Theor. 57 275401**]

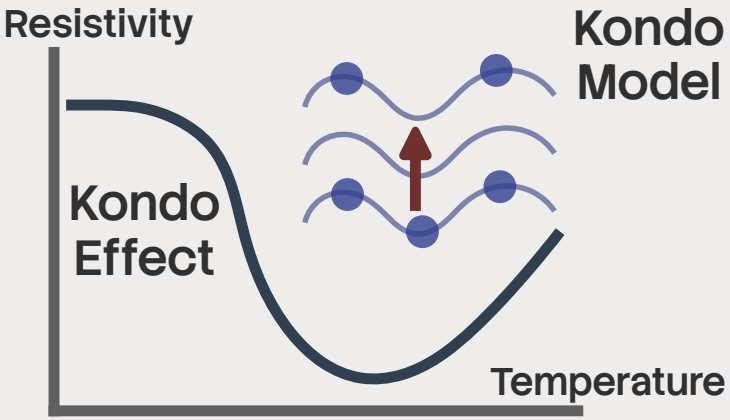
Chap 8. Conclusions

Introduction To Fermionic Criticality and Quantum Materials

What Do Condensed Matter Theorists Do?

Explain Experiments

- Build and solve theoretical models for experiments
- Compute measurables to explain observations
- Kondo effect, quantum Hall effect, superconductivity



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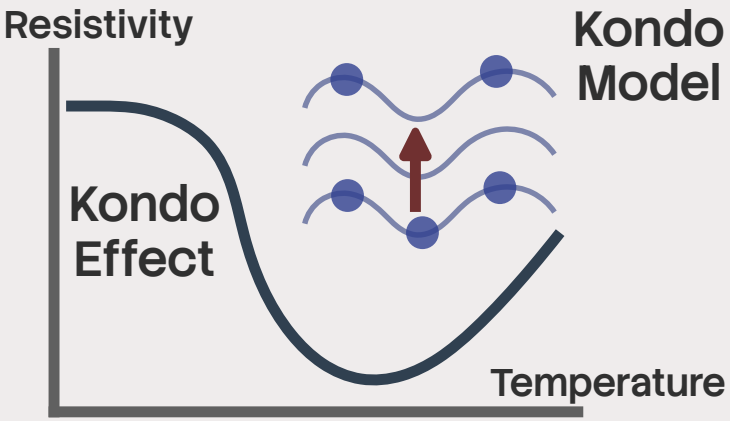
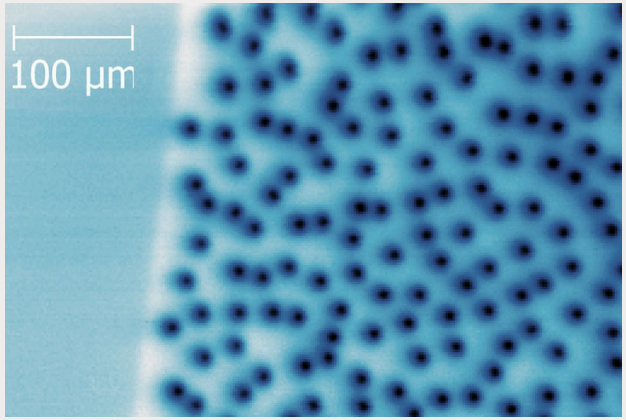


Predict Experiments

- Identify interesting theoretical models for new systems
- Predict novel phenomena by tweaking models and parameters
- Abrikosov vortex lattice, spin liquids



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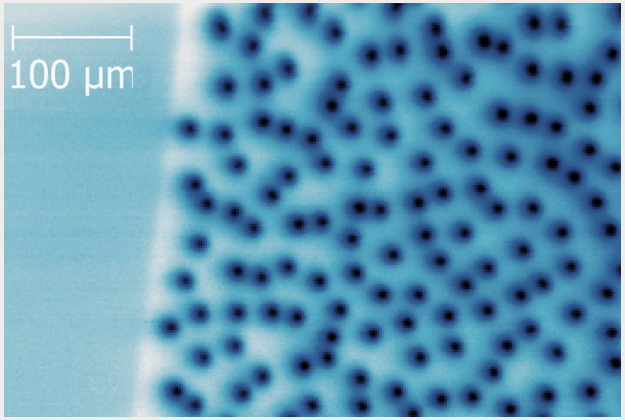

Classify experiments

- organize experimental signatures into universal principles
- come up with frameworks that explain a large class of phenomena
- Examples: Landau theory of phase transition

Quantum Materials

Class of materials that **cannot** be well-approximated by “classical” description

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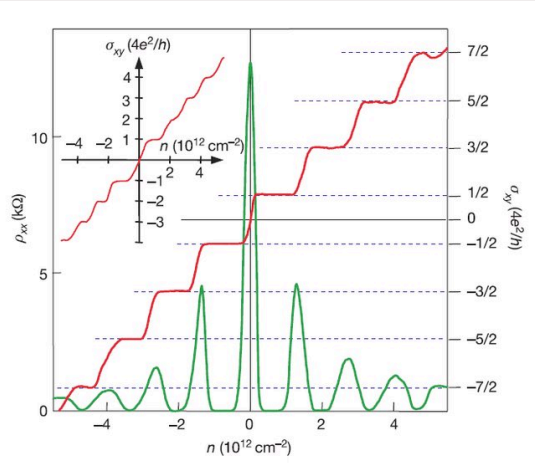

topological properties
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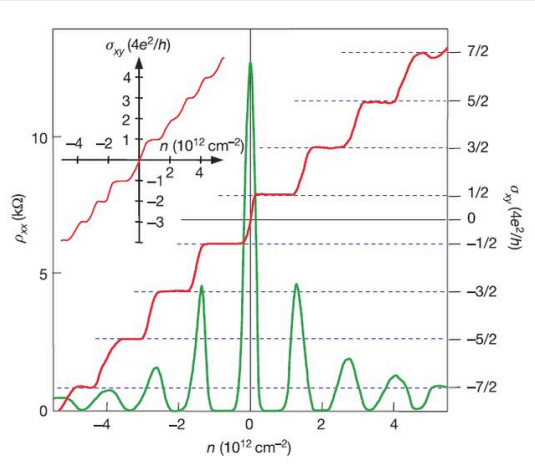

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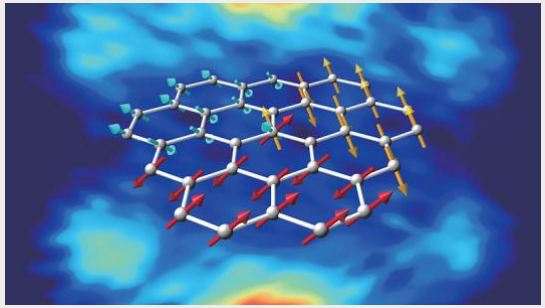

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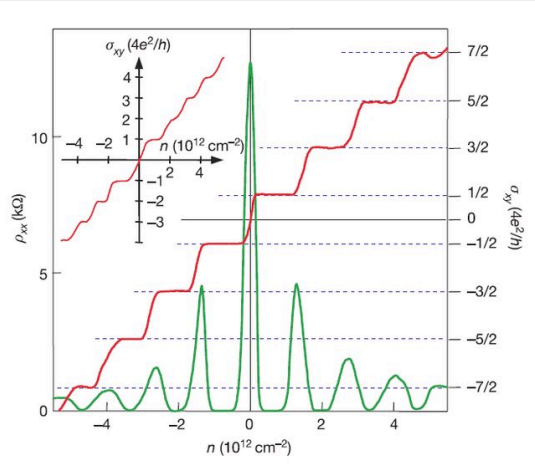

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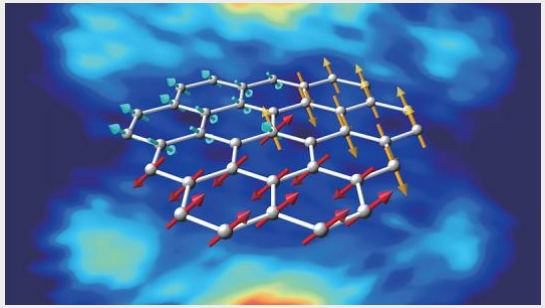

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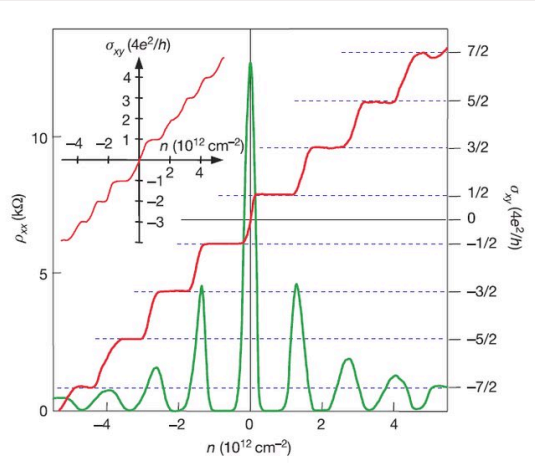

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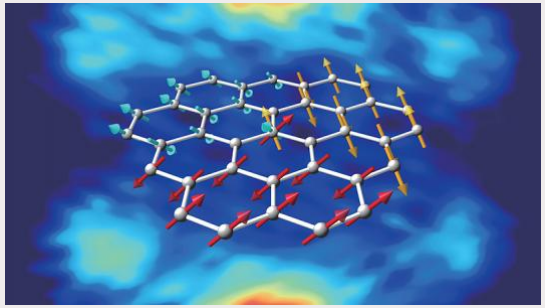

Needs dramatically **new ideas** and methods!

- **non-local/collective** order parameters for transition
- ability to work with a large number of **interacting particles**
- ability to work with **emergent** degrees of freedom

Strongly Correlated Phenomena

Materials with strong **local** correlation, **low dimensionality**, multiple electronic **orbitals**

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- Transition metal dichalcogenides (MoS_2)
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Localised *f* and *d* electrons **enhance** effect of correlations.

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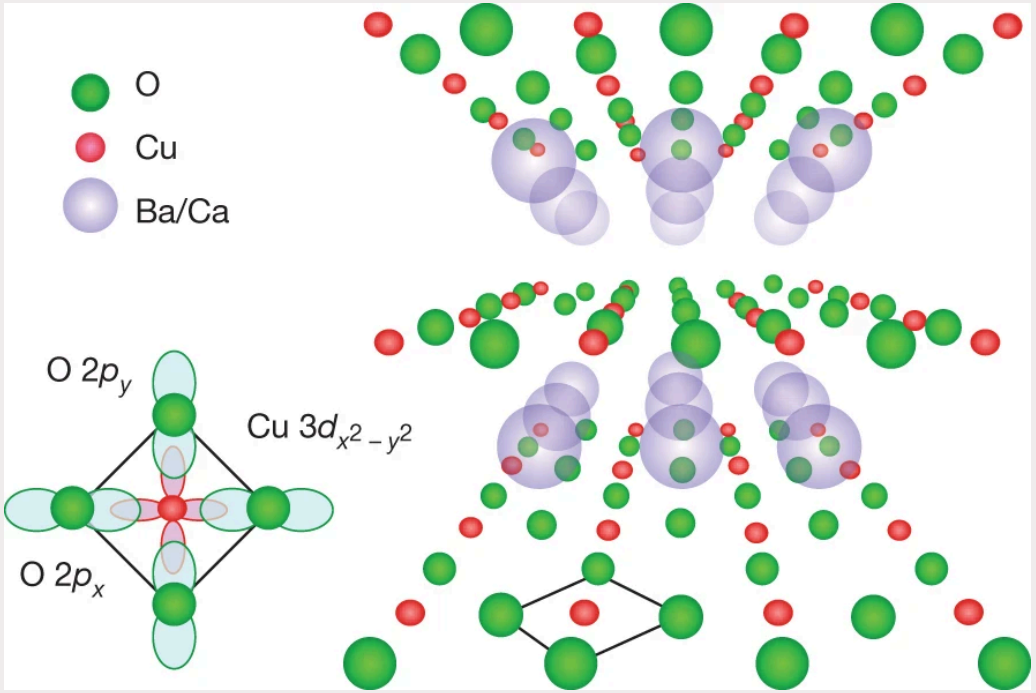

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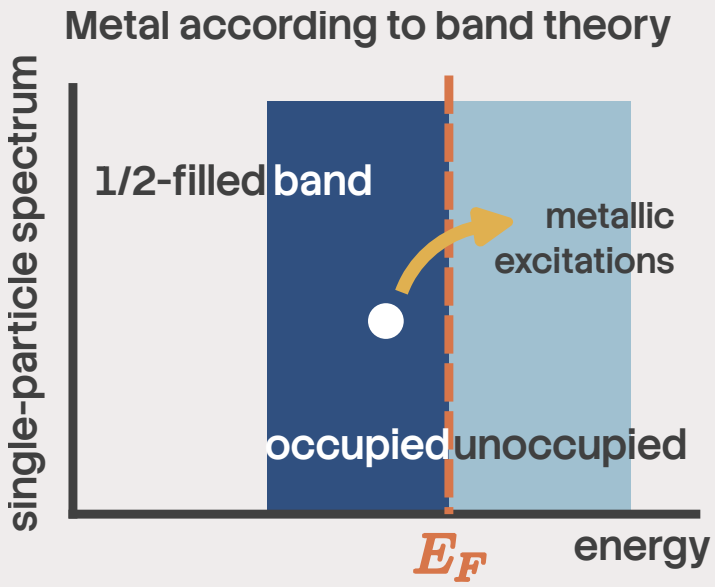

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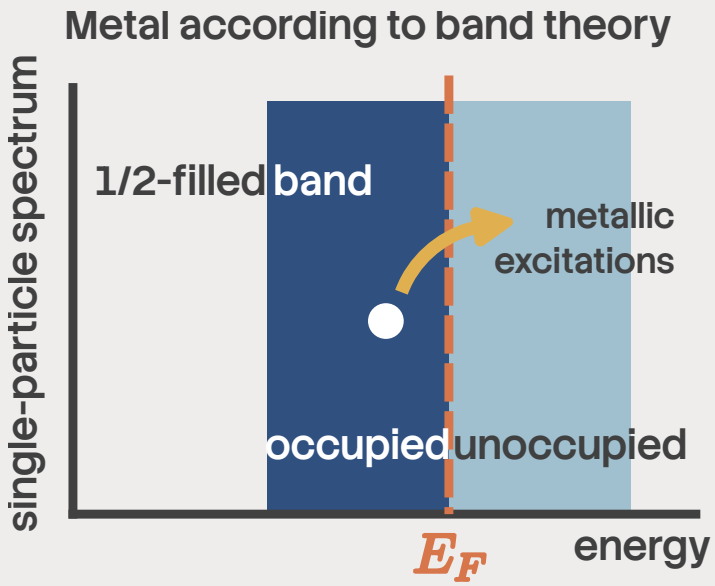

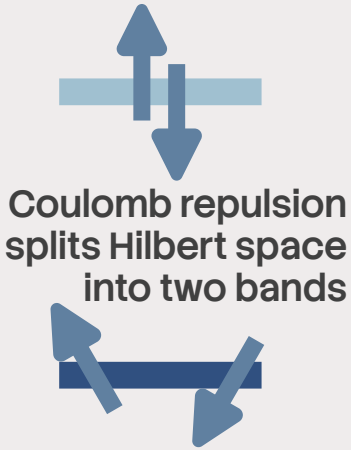

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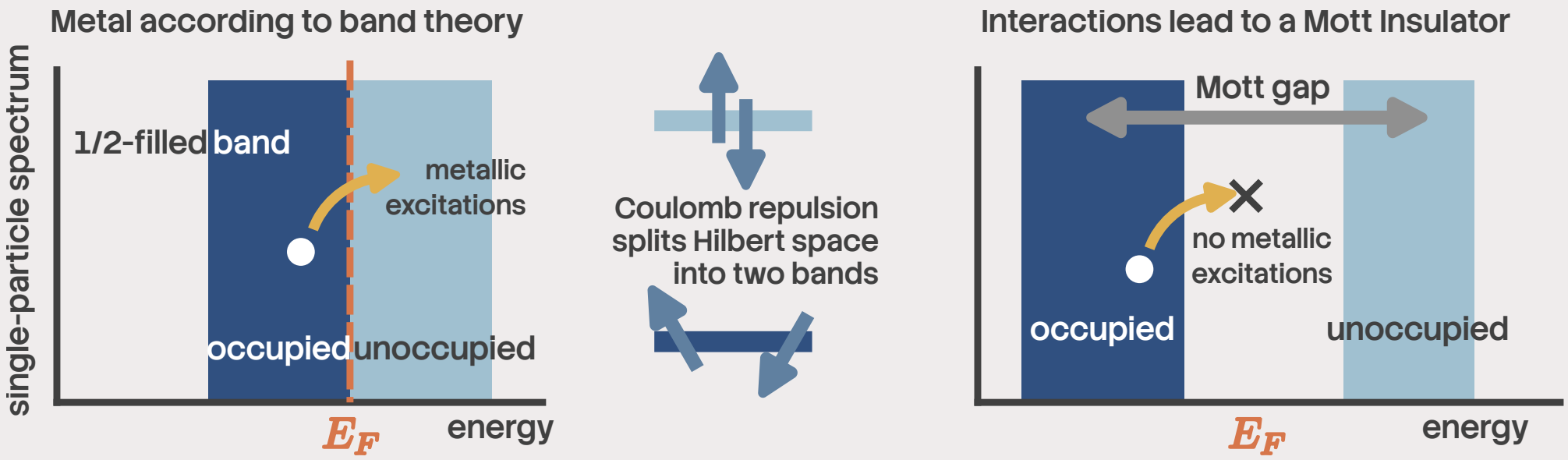

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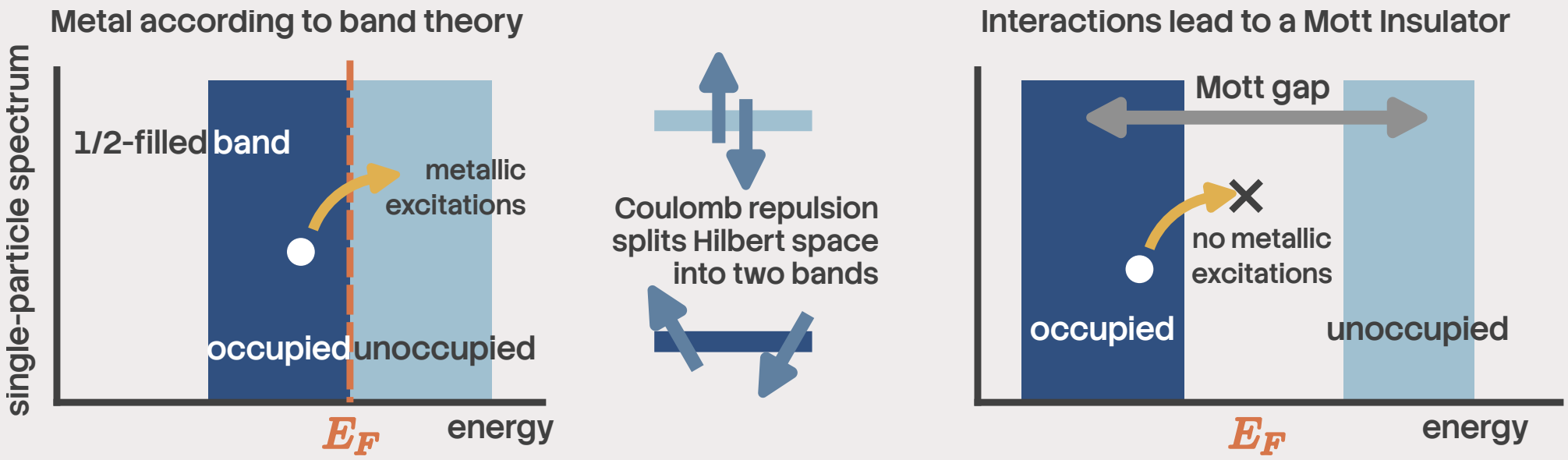

- Not accessible from 1-particle theory
- result of **manybody effects**

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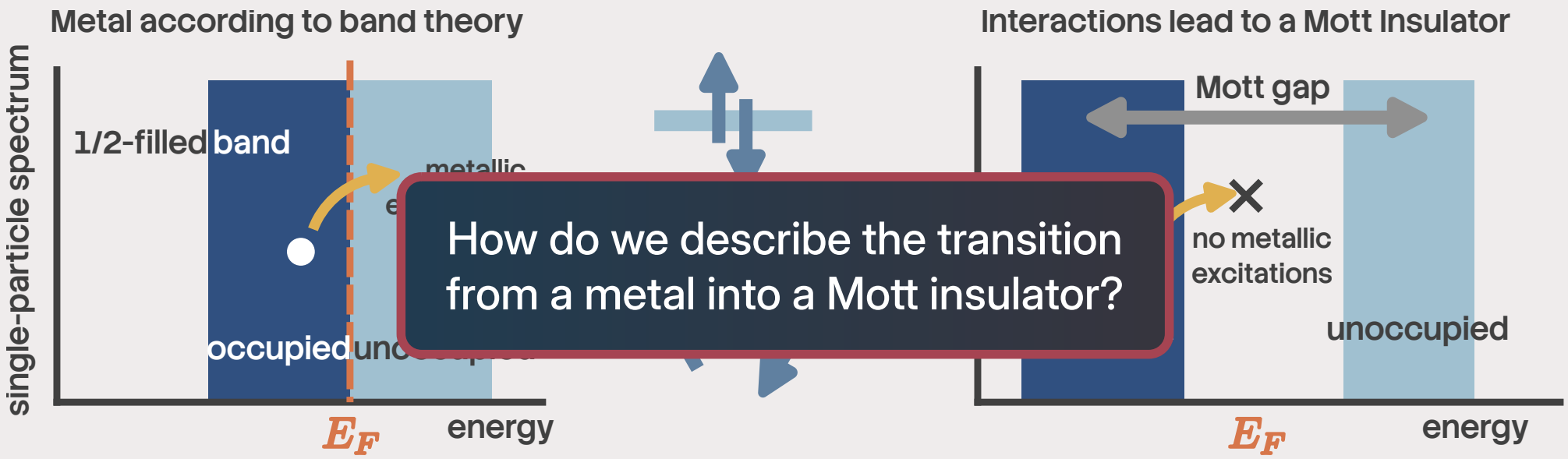

How do we describe the transition from a metal into a Mott insulator?

- Not accessible from 1-particle theory
- result of **manybody effects**

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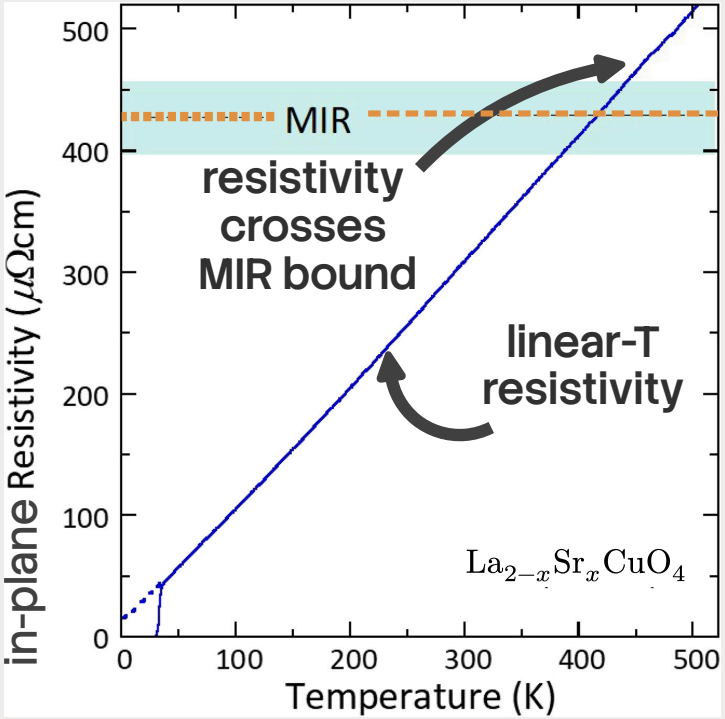
Phenomena: Strange Metallicity



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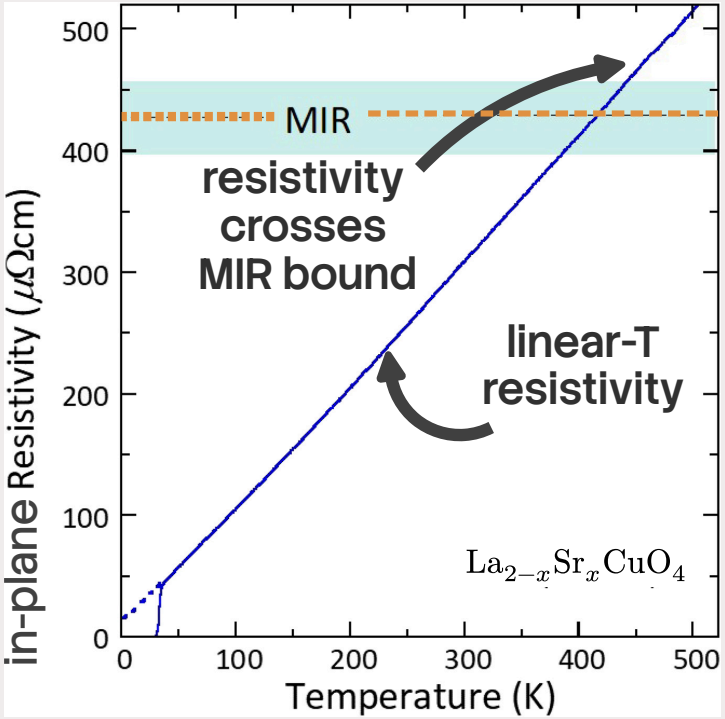
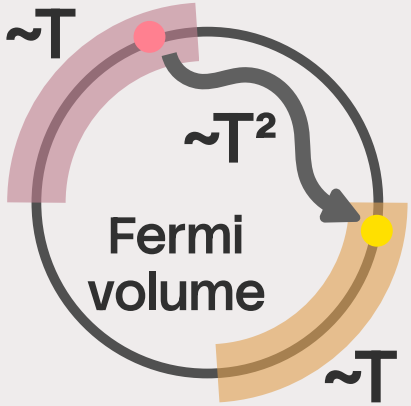


Conventional metallic theory predicts $R \propto T^2$



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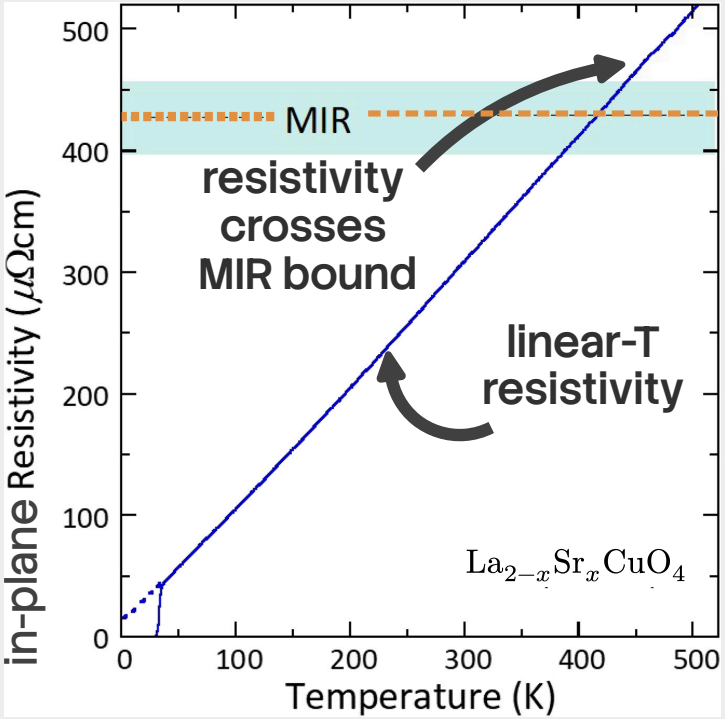

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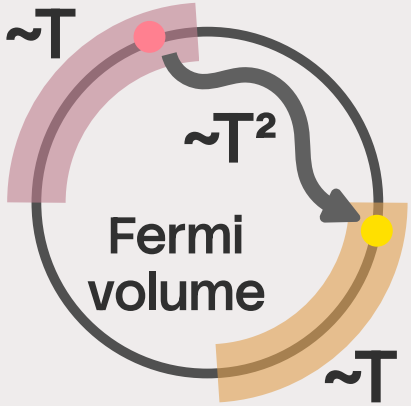
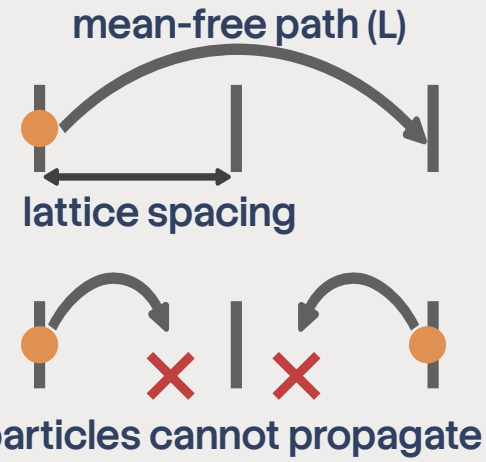


Source of $R \propto T^2$?

Single particles must propagate finite number of lattice spacing



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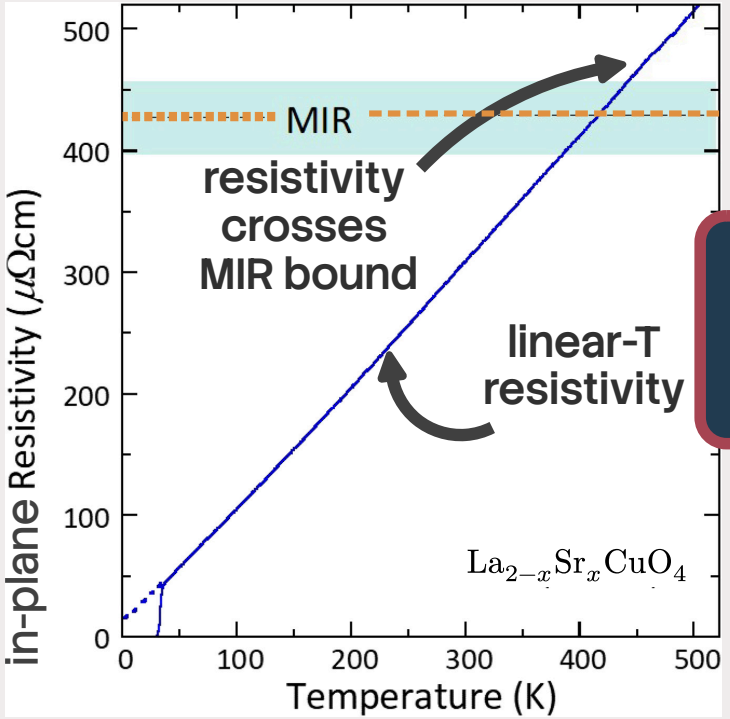

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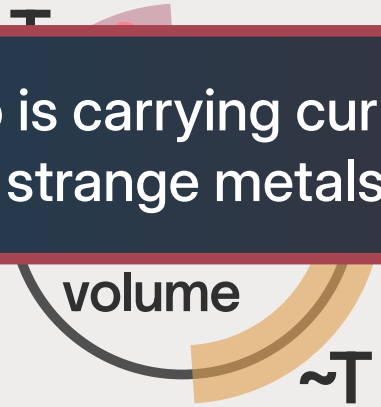
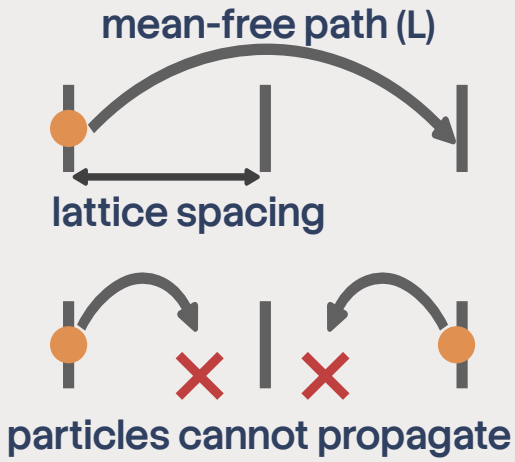


Source of $R \propto T^2$?

Who is carrying current in strange metals?

Single particles must propagate finite number of lattice spacing



Theoretical Models of Correlated Materials

Hubbard Model

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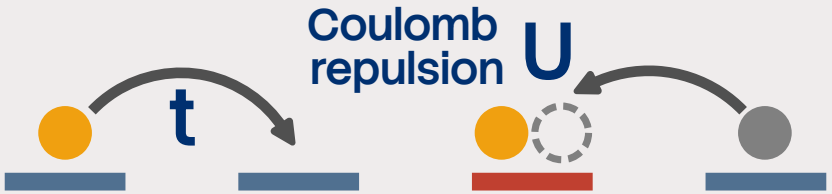

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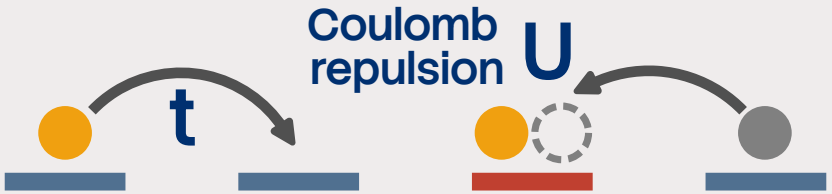

Hamiltonian: $-t \sum_{i,j,\sigma} (c_{i,\sigma}^{\dagger} c_{j,\sigma} + \text{h.c.}) + \sum_i U n_{i,\uparrow} n_{i,\downarrow}$

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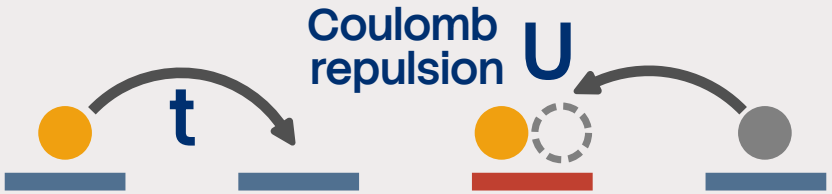

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- **Multi-orbital** model: one localised, the other delocalised

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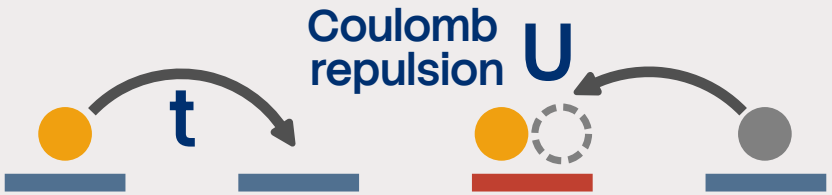

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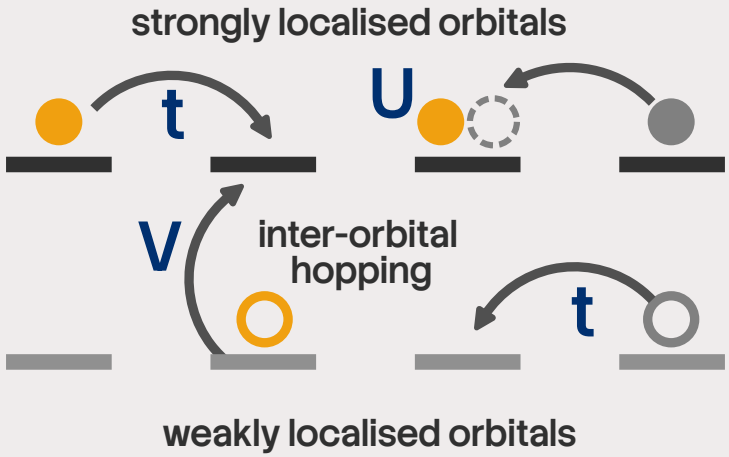

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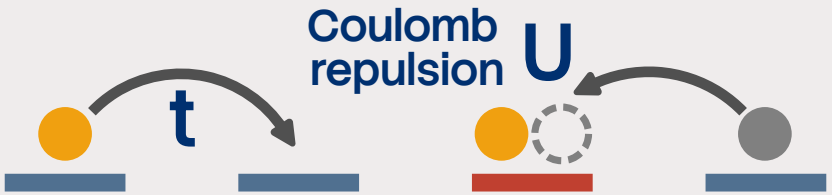

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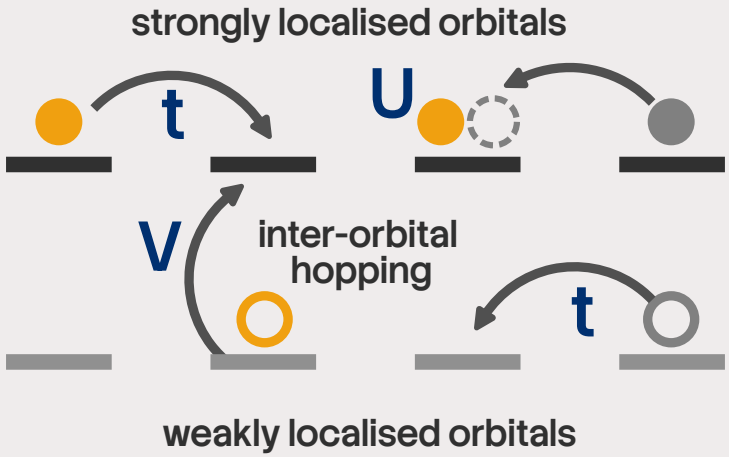

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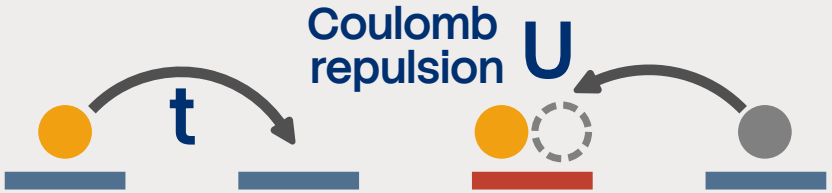

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$t \Longrightarrow$ Delocalisation
$U \Longrightarrow$ Localisation

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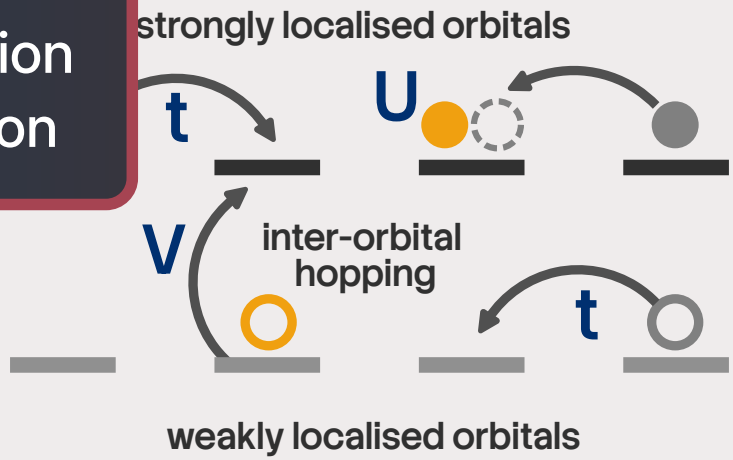

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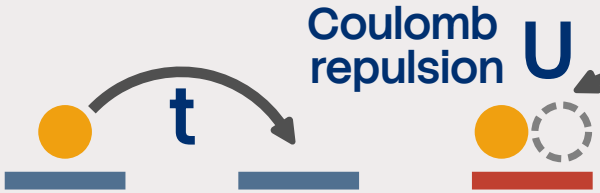
Hubbard Model

- Correlated electrons hopping on a lattice
- Hopping probability is t.

- Energy cost U if two occupy same site.



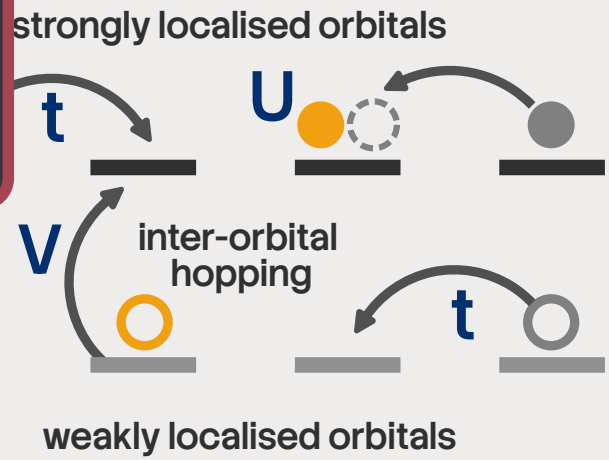
Hamiltonian: $-t \sum_{i,j,\sigma} (c^\dagger_{i,\sigma} c_{j,\sigma} + \sum_i U n_{i,\uparrow} n_{i,\downarrow}$

$t \Longrightarrow$ Delocalisation
$U \Longrightarrow$ Localisation

🔥 Competition
🫂 Correlations

Periodic Anderson Model

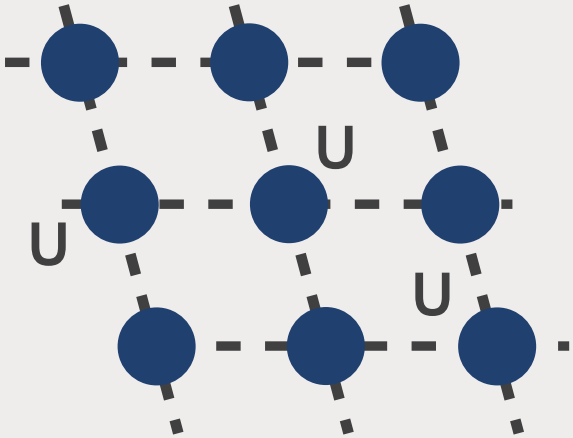
- **Multi-orbital** model: one localised, the other delocalised



s can hop across orbitals

Hamiltonian: $H_{\text{Hub}} + H_{\text{hop}} + V \sum_{i,\sigma} f^\dagger_{i,\sigma} c_{i,\sigma} +$ h.c.

Quantum Impurity Models as Auxiliary Models



Lattice models of correlated electrons are **hard to solve** directly.

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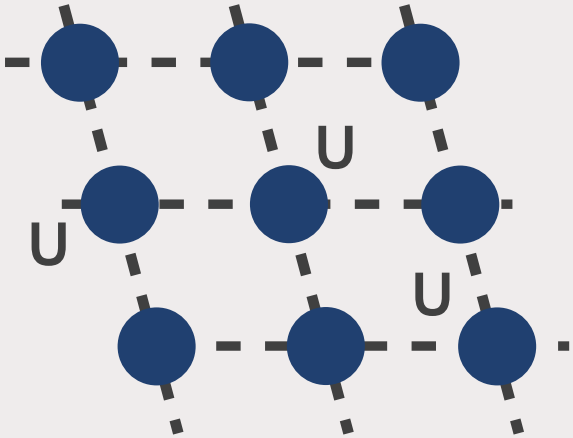

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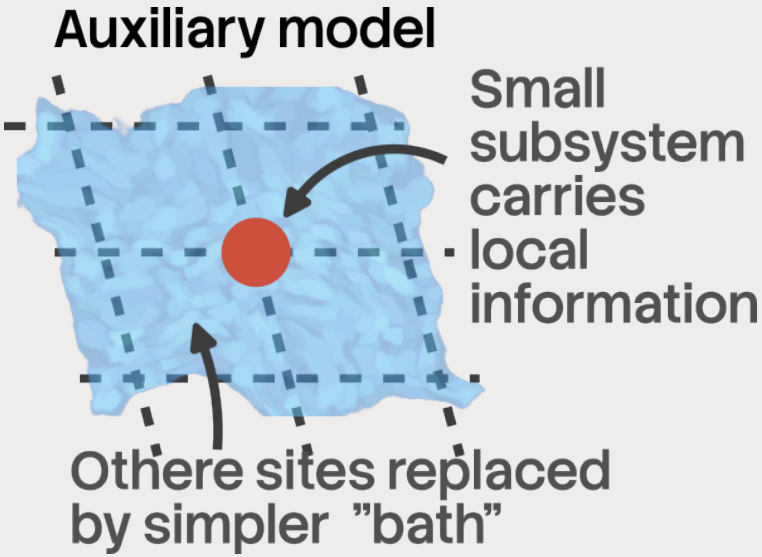

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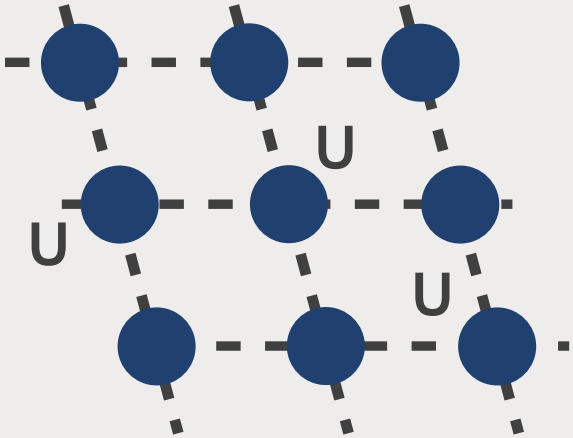

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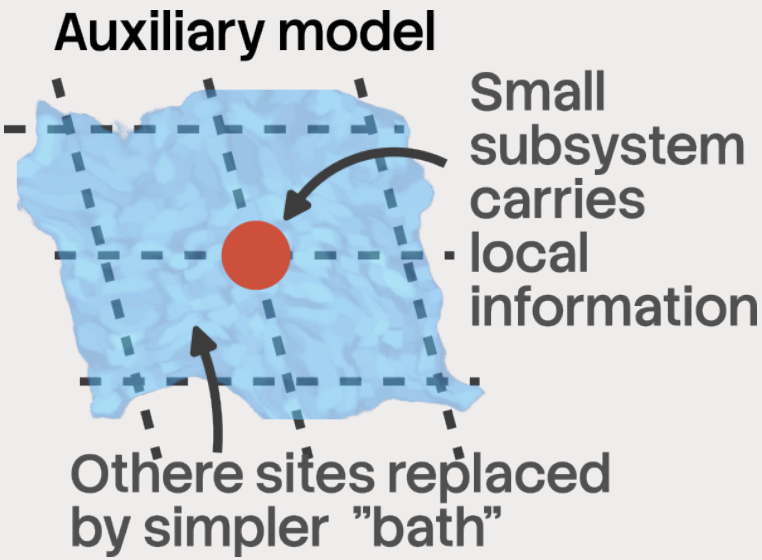

Example: Mean-field solution of Ising model

$$H = \sum_{i,j} J_{i,j} S_i^z S_j^z$$

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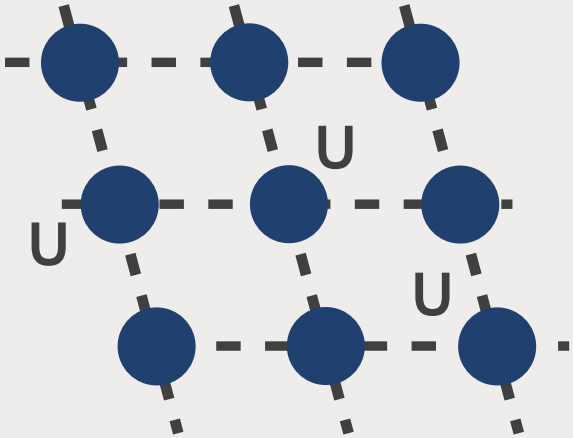

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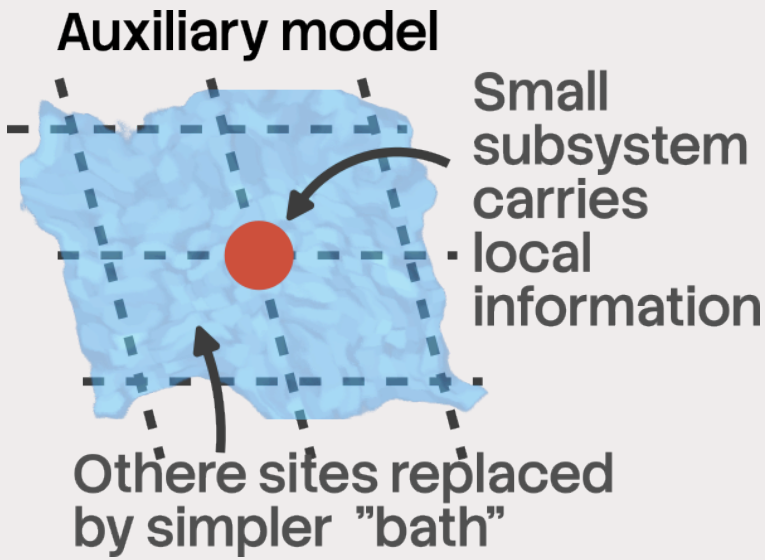

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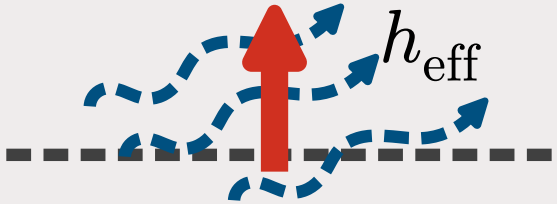
Mean-field approximation:

$$H = \sum_i h_{\text{eff}} S_i^z$$



Rest $\Longrightarrow$ effective field

Quantum Impurity Models as Auxiliary Models

Auxiliary Model for Interacting Quantum Systems?

Quantum Impurity Models as Auxiliary Models

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Ising model $\Longrightarrow$ **Classical**

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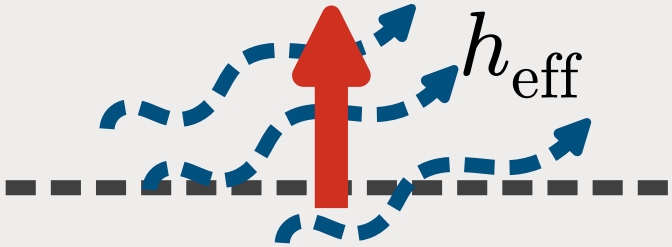
Ising model ⟹ **Classical**



Mean-field ⟹ **Static Model**

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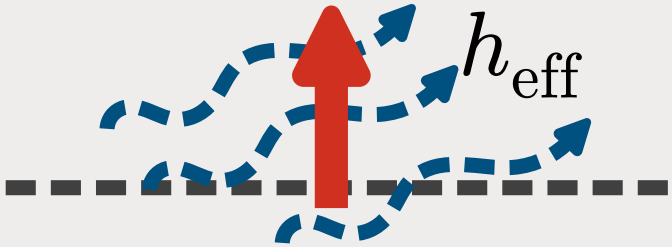

Mean-field ⟹ **Static Model**

Interacting Quantum Systems have **time dynamics**

Auxiliary Model has to be **interacting** in order to capture temporal dynamics!

Quantum Impurity Models as Auxiliary Models

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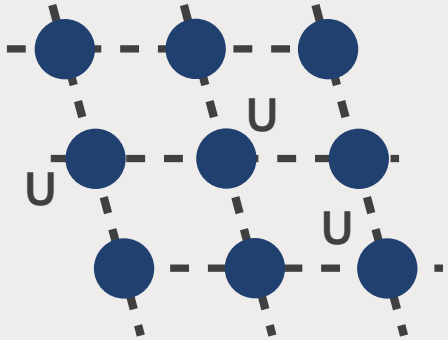
What is an appropriate auxiliary problem for lattice models with local correlation?



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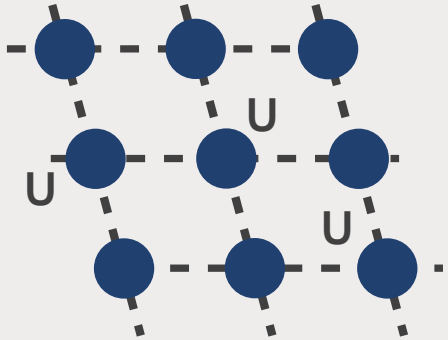

Insight from Limit of Infinite Dimensions!

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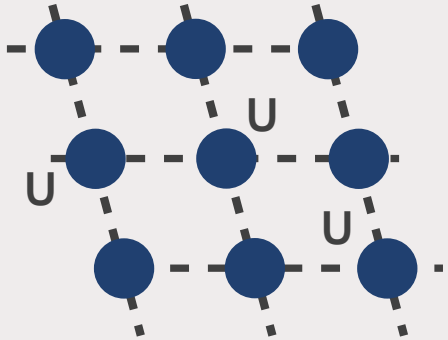
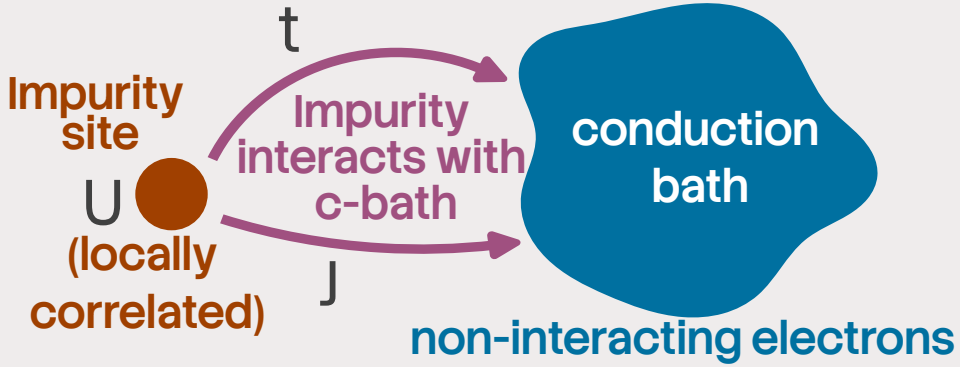


Insight from Limit of Infinite Dimensions!

Local physics ⟹ **Quantum Impurity Models**

Core Idea of Dynamical Mean-Field Theory



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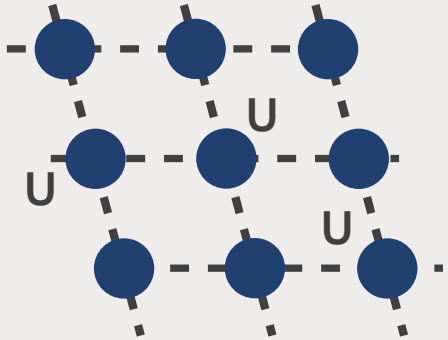

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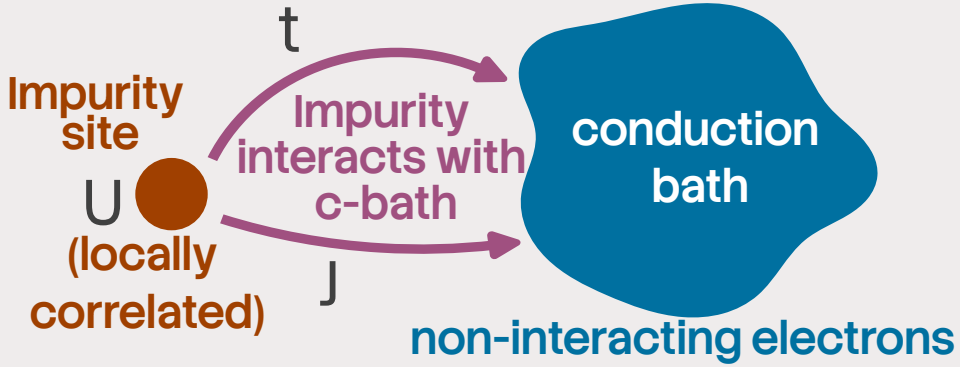

- Tractable
- Contains dynamics

(ω –dependence of correlations)

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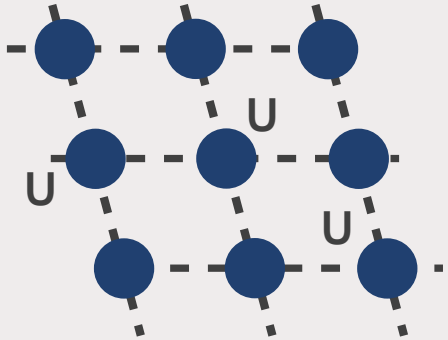

Local physics of correlated lattice models can be mapped on to impurity model.

Insight from Limit of Inf

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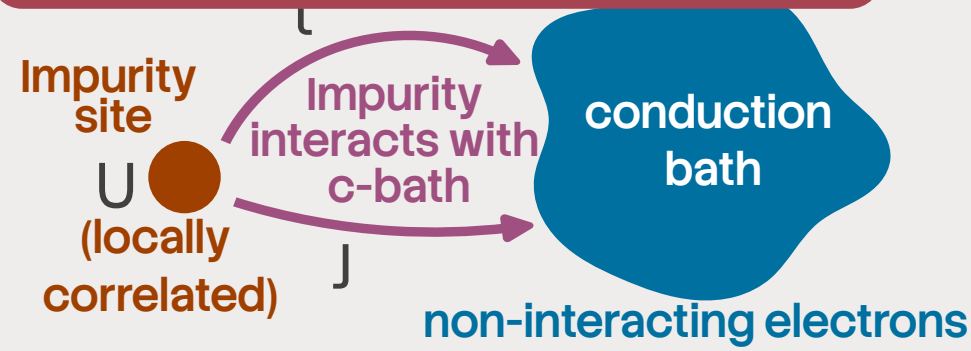

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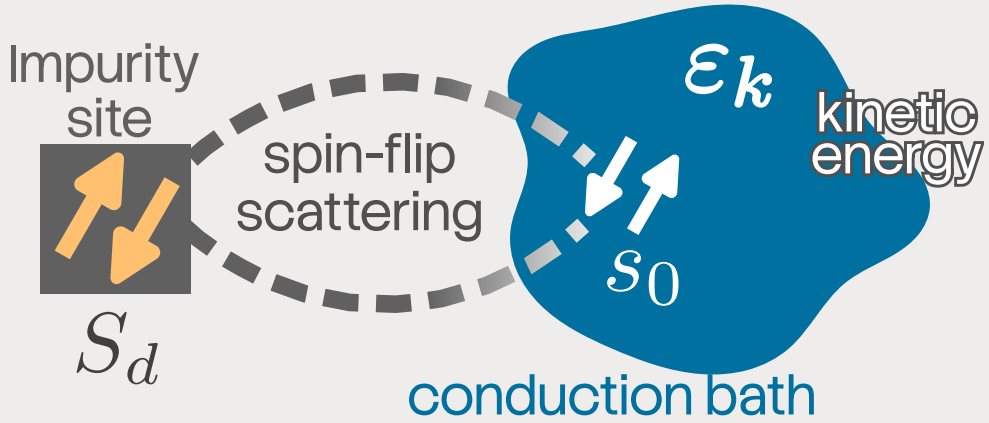
Phenomenology of Quantum Impurity Models

Kondo Model

- Impurity spin S_d + conduction bath
- Spin-flip scattering

$$H = \sum_{k,\sigma} \varepsilon_k n_{k,\sigma} + \frac{1}{4} J S_d^z s_0^z + \frac{1}{2} S_d^+ s_0^- + \frac{1}{2} S_d^- s_0^+$$



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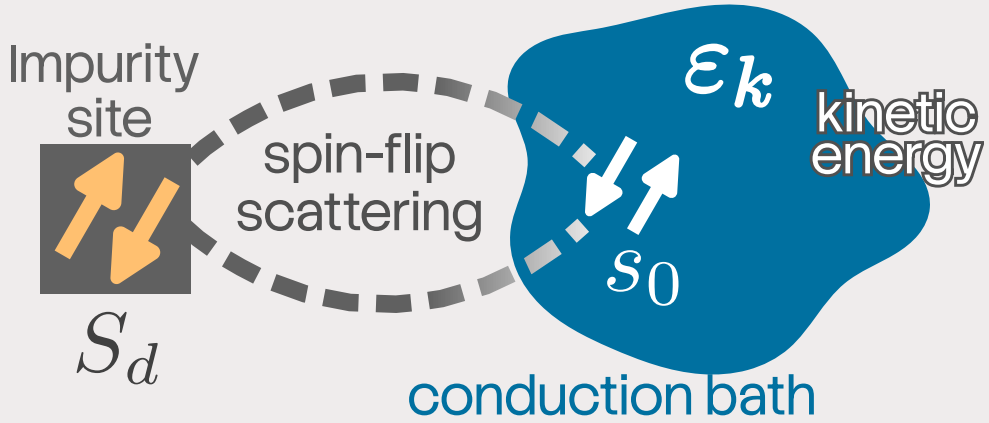

Low-energy ($T \ll 1$) Properties: Kondo Screening

- Kondo coupling generates **entanglement**
- Impurity DOS supports **gapless excitations**
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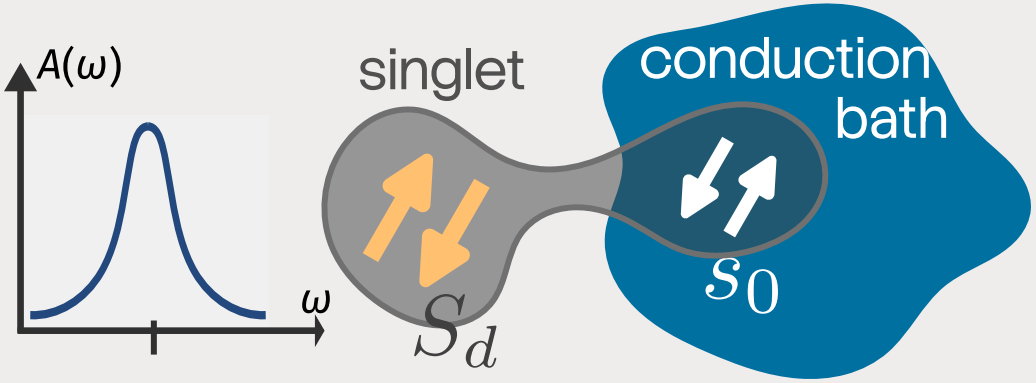

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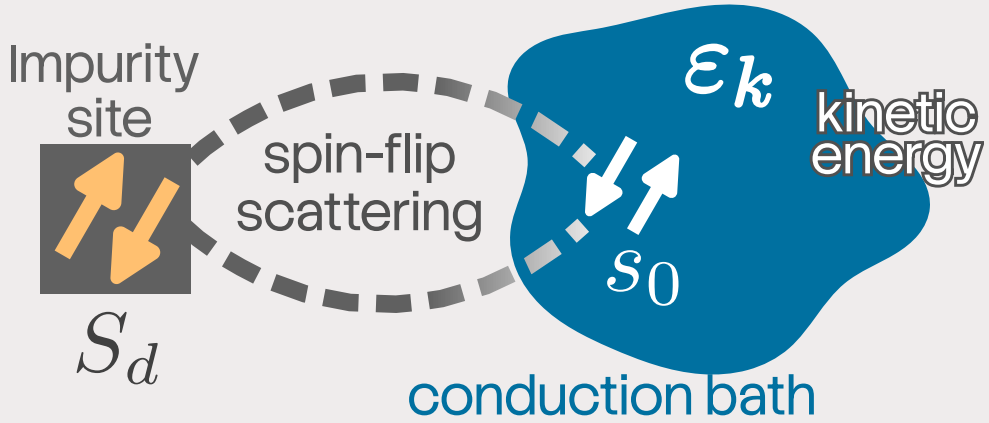

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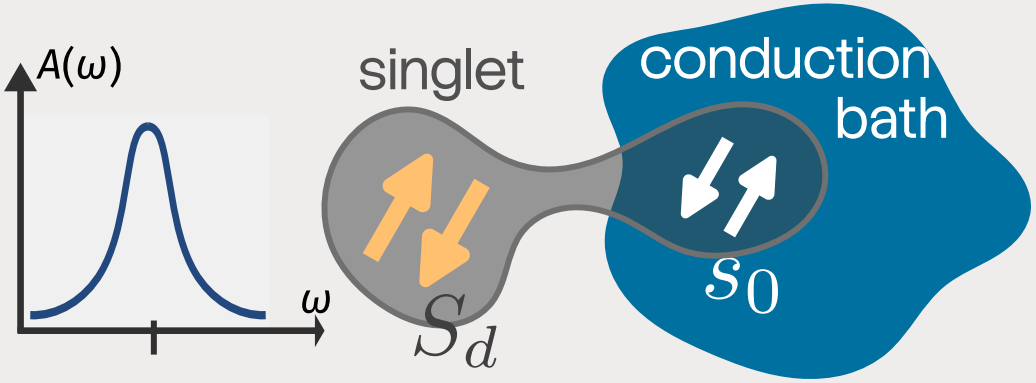
$\Longrightarrow$ **metallic state** in auxiliary model language.



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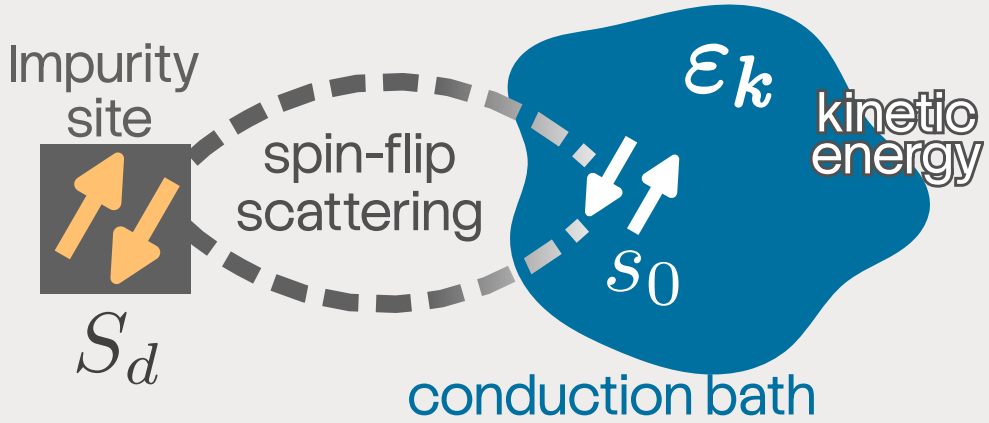
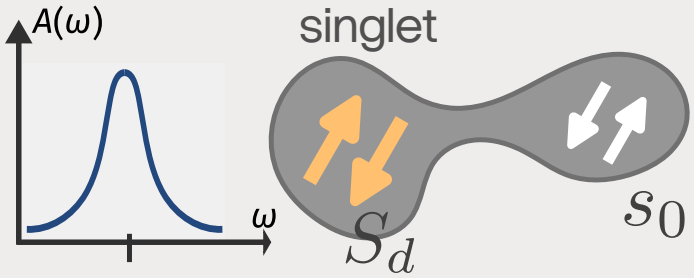


Metallic State



How Do We Get an Insulating State?

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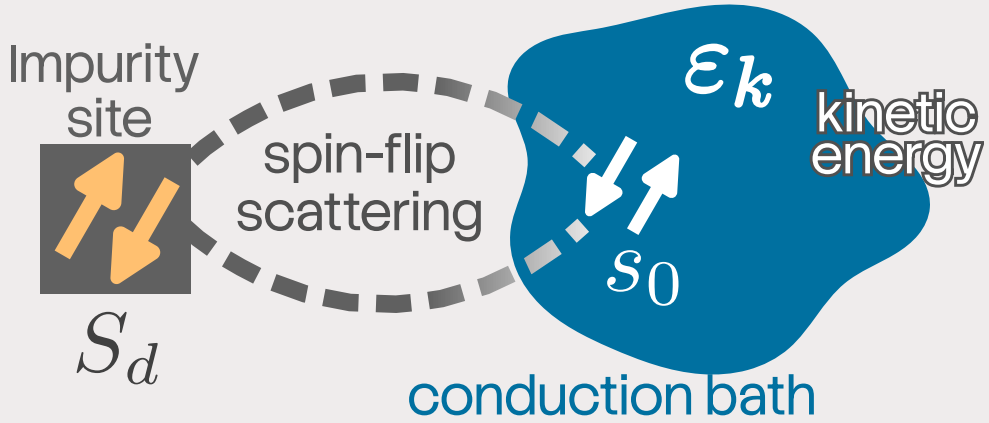

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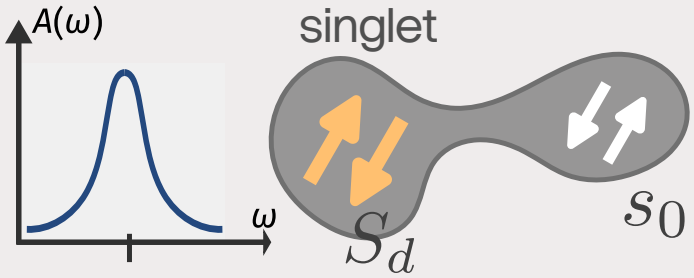
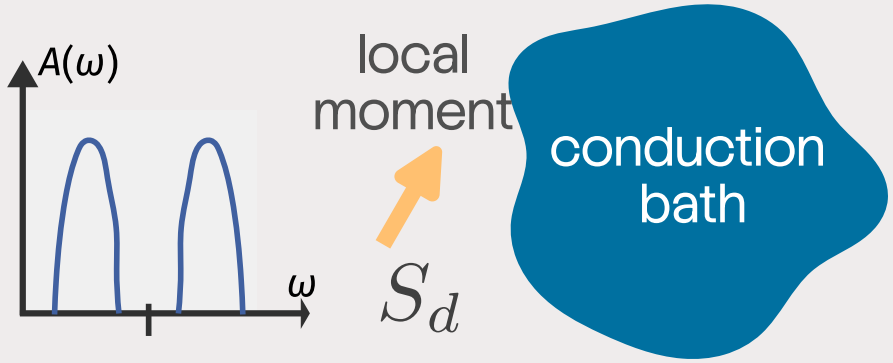


How Do We Get an Insulating State?

- Tweak the bath
- Destroy screening
- Local moment emerges



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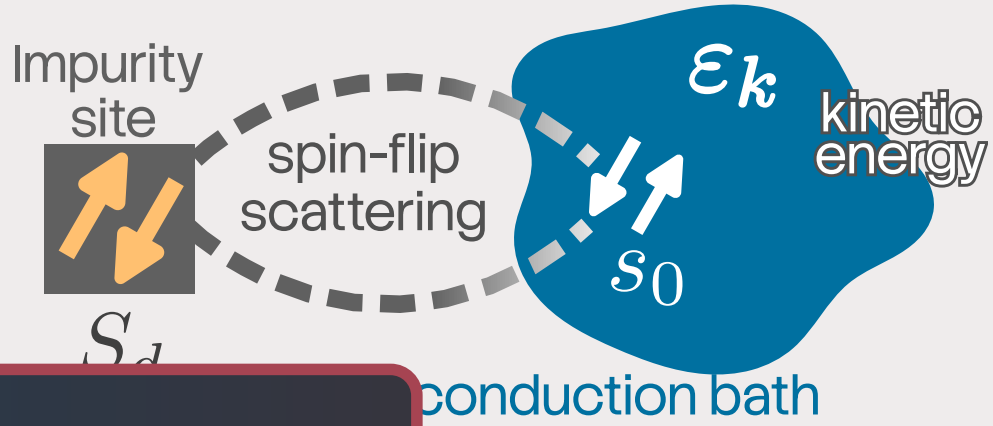
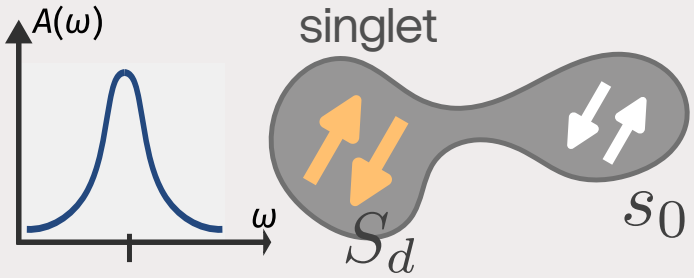


Kondo screening ⟹ Metallicity

Local moment ⟹ Mott insulation

Metallic State



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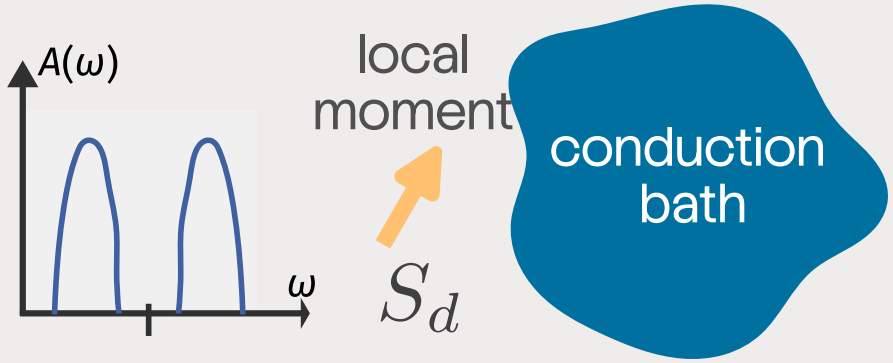

So, What's Left To Do?

- Questions about solving models
- Questions about developing methods

Minimal Model For Kondo Breakdown

A Route to Mott Localisation in Infinite Dimensions

New J. Phys. 25 113011 (2023)
AM, N S Vidhyadhiraja, A Taraphder, S Lal

Capturing Spatial Fluctuations and Lattice Geometry

Impurity-Lattice Mapping Beyond Mean-Field

New J. Phys. 28 013503 (2026)
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Kondo Breakdown On The Square Lattice

Revisiting the Mott Transition in Two Dimensions

New J. Phys. 28 013503 (2026)
AM, S R. Hassan, A Mukherjee, N S. Vidhyadhiraja, A Taraphder, S Lal

Competing tendencies In a Multi-Orbital System

Kondo screening vs. Mott localisation in a Heavy-Fermion model

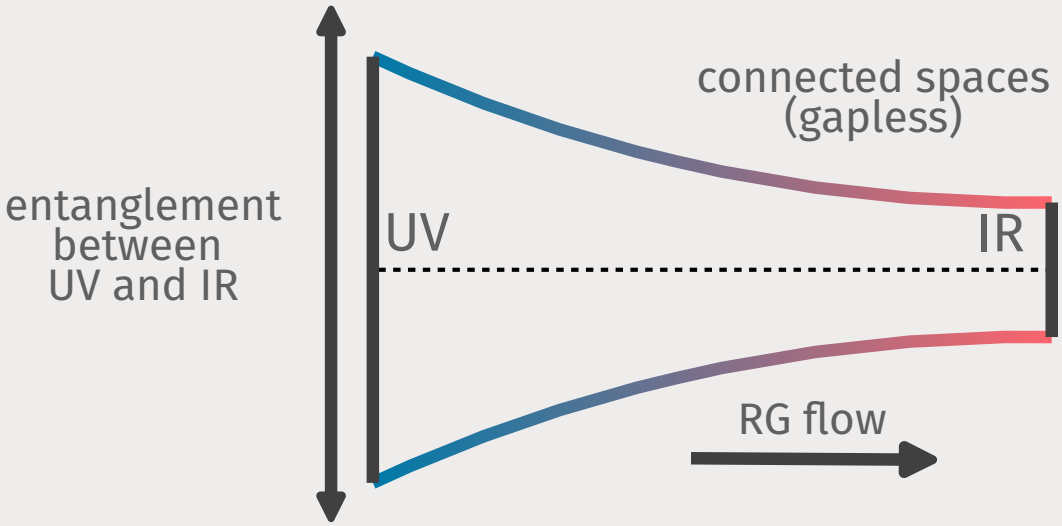
(in progress)

AM, S Lal

Holography of Entanglement in 2D Free Fermions

Emergent Geometry and Fermionic Criticality

J. Phys. A: Math. Theor. 57 275401 (2024)
AM, S Patra, S Lal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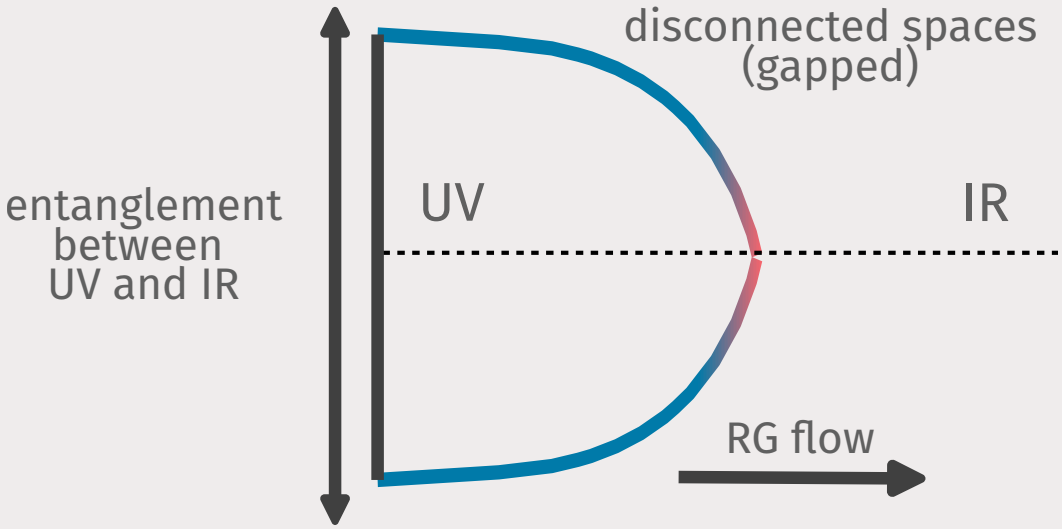
UV
IR
RG flow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Closing Remarks

Summary of Primary Results

Main Conclusions and Takeaways

Future Work